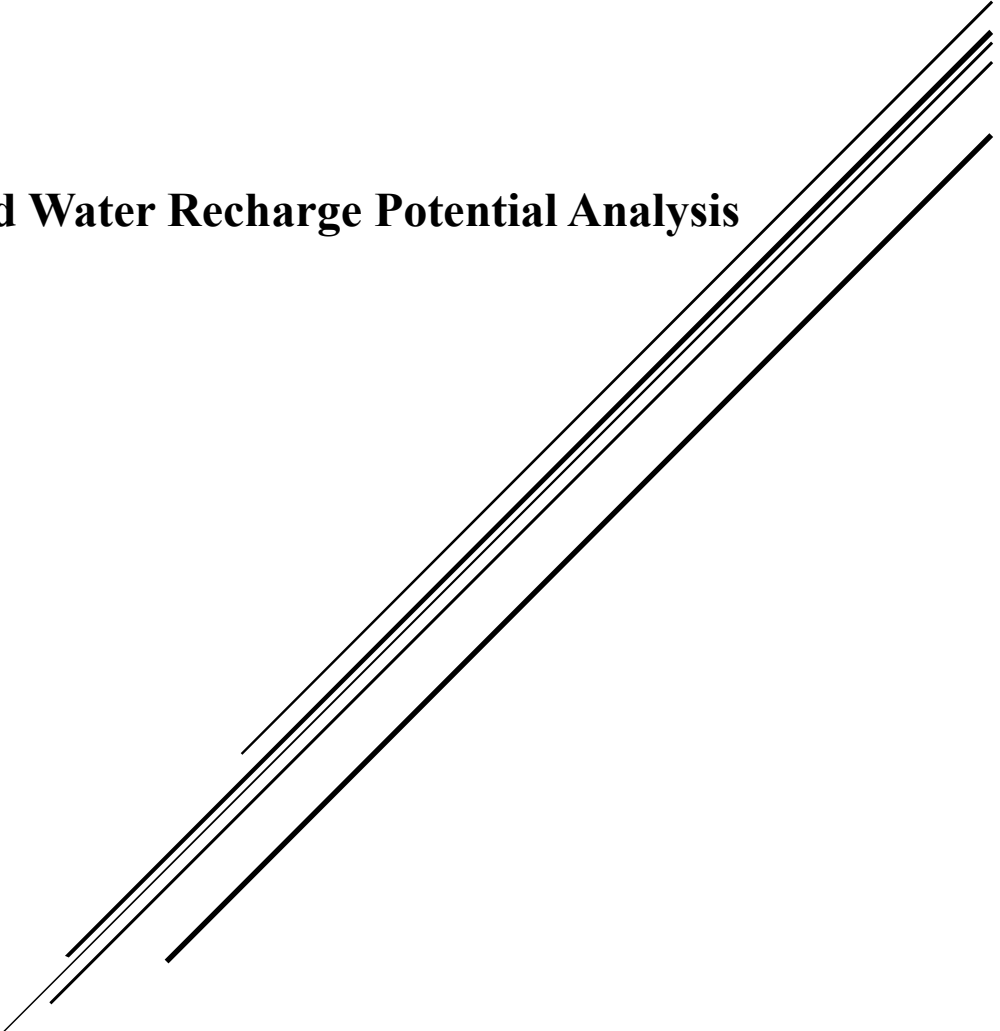


PL2408 - Remote Sensing for Planning

Assignment

Ground Water Recharge Potential Analysis



- Shahidh M.S
- 212360F
- Department of Town & Country Planning
- Faculty of Architecture
- University of Moratuwa

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1. Background and Challenges

The application of groundwater recharge potential within the Kurunegala District stands as a critical endeavor given its significance in sustaining local water resources and ecosystem resilience. Kurunegala is an agriculturally vibrant region, which is heavily depend on groundwater for irrigation, domestic use, and industrial purposes. The sustainability and reliability of these groundwater sources face continue threats because of rapid urbanization, changing land use patterns, and the impacts of climate change.

Groundwater recharge potential analysis holds a significant role in sustainable water resource management. By evaluating areas conducive to groundwater replenishment, it aids in identifying regions where infiltration and natural processes contribute most to aquifer recharge. This analysis guides land-use planning, enabling the protection of critical recharge zones and mitigating contamination risks. Understanding recharge potential assists in designing effective groundwater conservation strategies, ensuring long-term water availability for people, agriculture, and ecosystems.

Main challenge addressed by this analysis lies in understanding and quantifying the factors influencing groundwater recharge rates within the Kurunegala District. There is no documented data available regarding groundwater in the Kurunegala district, including well locations and groundwater depth information. And, in this analysis there is no direct results to get information about Ground water recharge potential. The analysis aims to provide insights into mitigating the adverse effects of over-extraction and contamination of groundwater resources, ensuring their long-term viability.

2. Role of Remote Sensing

Remote sensing technology plays a major role in addressing the challenges associated with evaluating groundwater recharge potential in the Kurunegala District. Using Landsat 8 bands and ArcGIS, remote sensing offers a comprehensive and efficient means to collect, analyze, and interpret the data for land cover, soil properties, vegetation health, and hydrological characteristics across large spatial extents.

By utilizing multispectral bands captured by Landsat 8, the analysis incorporates various parameters like land use/land cover changes and vegetation indices. These elements serve as key indicators in identifying potential areas for groundwater recharge, delineating areas with higher infiltration rates, and assessing the influence of land use dynamics on recharge patterns.

Remote sensing techniques enable the integration of diverse datasets, including information obtained from research articles and waterboard reports, thereby enhancing the accuracy and reliability of the analysis. The ability to monitor changes over time aids in understanding temporal variations in recharge potential, facilitating informed decision-making for sustainable water resource management strategies.

3. Key findings

Ground water recharge potential analysis was performed to Kurunegala district using ArcMap and Landsat 8 bands for 2022. There is no direct method for this analysis. For this we have to do several analyses and interpret those results. Such as Land use Classification, vegetation analysis, hydrological modelling etc.

3.1 Land use Classification

In the context of Kurunegala District performing land use classification is essential in groundwater recharge potential analysis because it helps in identifying and understanding the distribution and characteristics of various land cover types within the study area. Different land use types, such as vegetation, Buildup area, and water bodies, directly impact the infiltration rates. For instance, vegetation can enhance infiltration by promoting soil porosity and reducing runoff, while Buildup area, like buildings and pavements have low infiltration, leading to increased surface runoff. Understanding these land cover types allows for better assessment of the potential for water infiltration, recharge rates, and overall water availability for groundwater replenishment. In the land use map of the Kurunegala district below, it shows that the majority of the study area is covered by built-up areas (shown in red).

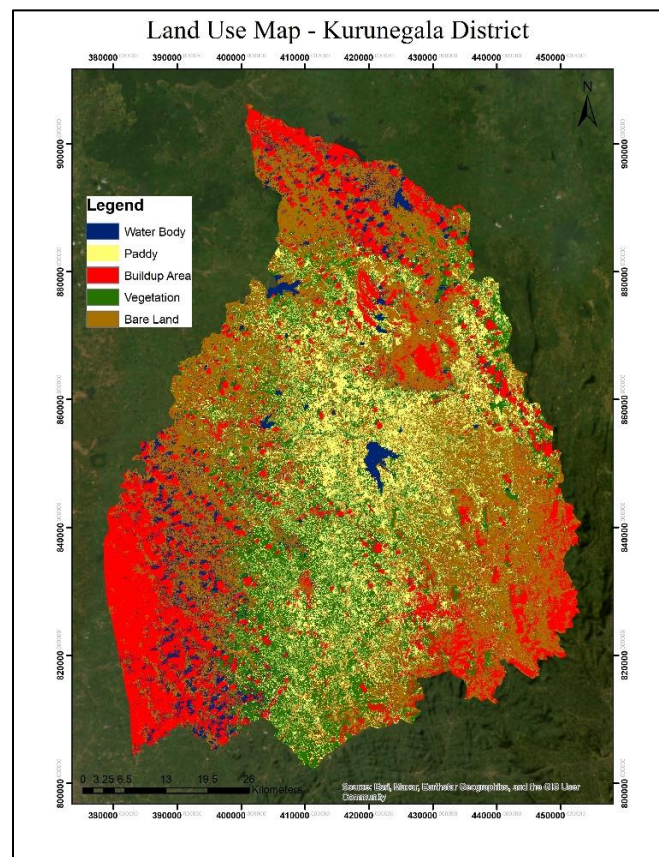


Figure 1 - Land use Map

3.2 NDVI vegetation analysis

Calculating NDVI using Landsat data in ArcGIS helps us understand how healthy and dense vegetation is in an area. Healthy plants reflect more NIR and absorb more Red light, resulting in higher NDVI values. This data tells us about the vegetation cover, which directly affects how water moves through an area. When plants are healthy and plentiful, they help water soak into the ground instead of running off the surface. Understanding NDVI helps us figure out how plants impact water movement, which is crucial for ensuring sustainable water management and a thriving environment.

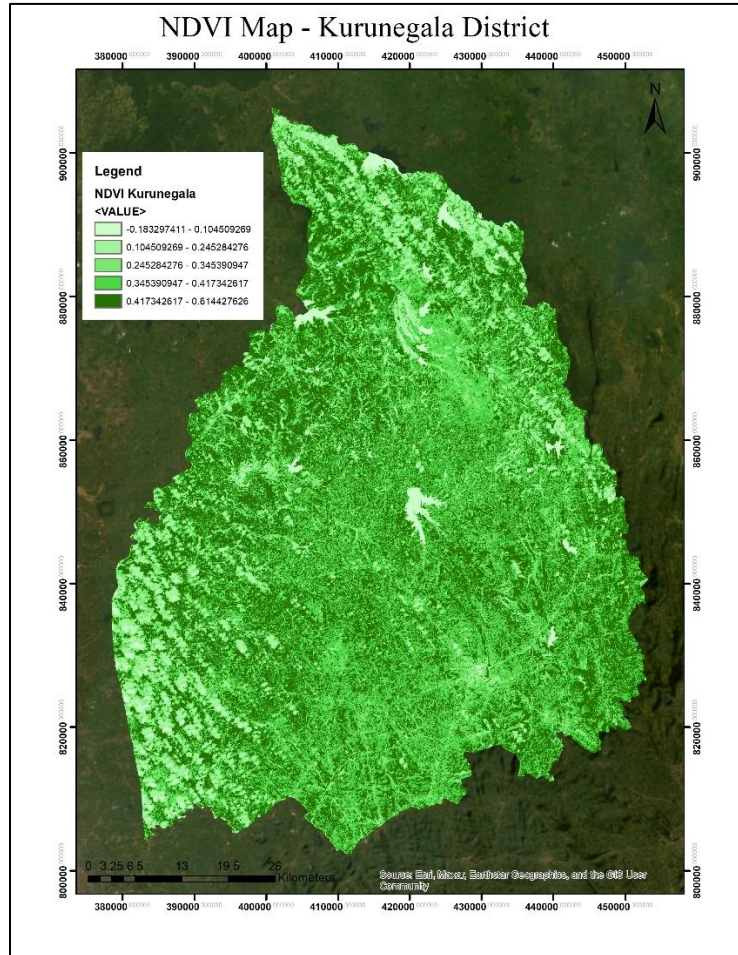


Figure 2 - NDVI Map

Based on above NDVI map of Kurunegala district, the groundwater recharge potential in the Kurunegala district is

- Very high: The areas in green have NDVI values above 0.417342617. These areas are likely to have the highest rates of groundwater recharge, due to the dense vegetation cover.

- High: The areas in light green have NDVI values between 0.345390947 and 0.417342617. These areas also have a good potential for groundwater recharge.
- Moderate: The areas in yellow have NDVI values between 0.245284276 and 0.345390947. These areas have a moderate potential for groundwater recharge.
- Low: The areas in orange have NDVI values between 0.104509269 and 0.245284276. These areas have a low potential for groundwater recharge.
- Very low: The areas in red have NDVI values below 0.104509269. These areas are likely to have the lowest rates of groundwater recharge, due to the sparse vegetation cover.

3.3 Hydrological Modelling (Soil Conservation Service)

Estimating how water infiltrates the soil and runs off the surface using hydrological models like the Soil Conservation Service (SCS) Curve Number method in ArcGIS is important for this application. This analysis is crucial for understand the groundwater pattern. By studying how things like the kind of land cover and soil affect how water moves, we can find places where more water can get into the ground.

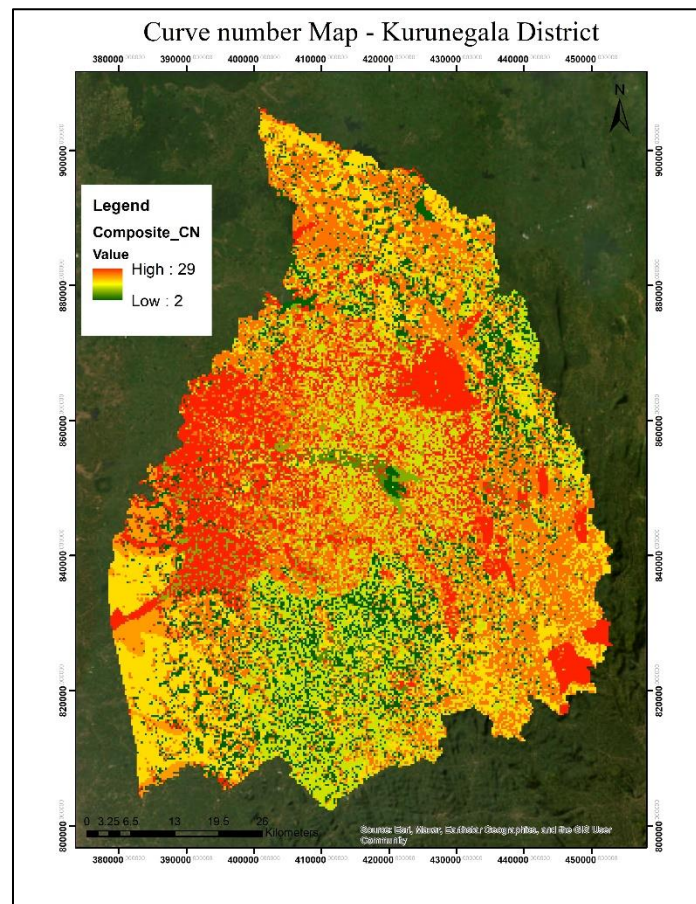


Figure 3 - SCS Curve number Map

Performing the weighted combination of land cover and soil types to calculate the composite Curve Number using ArcGIS tools facilitates the estimation of surface runoff potential and aids in hydrological modeling for watershed management and planning. Adjusting weights and validating against observed data improves the reliability of the derived CN values.

$$\text{SCS CN} = (\text{Weight_LandCover} * \text{LandCover_CN}) + (\text{Weight_SoilType} * \text{SoilType_HydroGroup})$$

The Curve Number (CN) values assigned to different land cover types are based on their hydrological characteristics, infiltration rates, and runoff potential. Lower values indicate lower runoff potential and higher infiltration. According to that in above map areas which are shown in red color have high infiltration rate.

3.4 Topography Analysis

Topography analysis is crucial in groundwater recharge potential analysis because it provides essential information about the land surface characteristics that influence the movement and distribution of water. By examining the topography, such as elevation, slope we can identify areas prone to runoff, infiltration, or groundwater recharge in Kurunegala district.

Slope Map

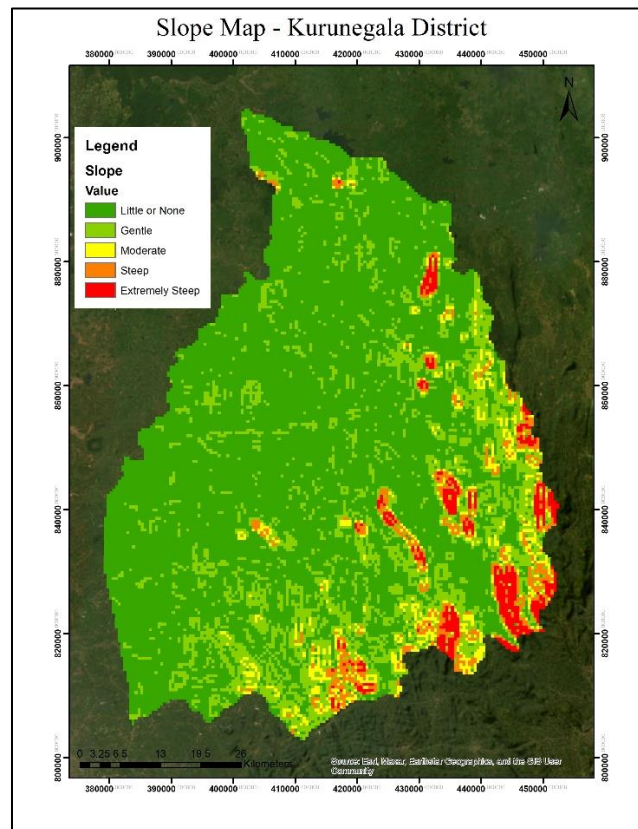


Figure 4 - Slope Map

The above map shows the slope gradient of the Kurunegala District. Steeper slopes are represented by darker colors, while gentler slopes are lighter. The legend indicates a range of slope values, from "Little or None" to "Extremely Steep."

Steeper slopes Generally have lower groundwater recharge potential. This is because rainwater tends to run off quickly on steep slopes, leaving less time for it to infiltrate into the ground and recharge the groundwater table. Gentler slopes have higher groundwater recharge potential. Rainwater has more time to infiltrate the soil and percolate downward to recharge the groundwater table.

- High recharge potential: The areas in light green and yellow, particularly in the central and northwestern parts of the district, have gentle slopes and are likely to have good groundwater recharge potential.
- Moderate recharge potential: The areas in light orange, mainly in the southern and eastern parts, have moderate slopes and may have moderate groundwater recharge potential.
- Low recharge potential: The areas in red, concentrated in the northeastern and southwestern corners, have steeper slopes and are likely to have low groundwater recharge potential.

Elevation Map

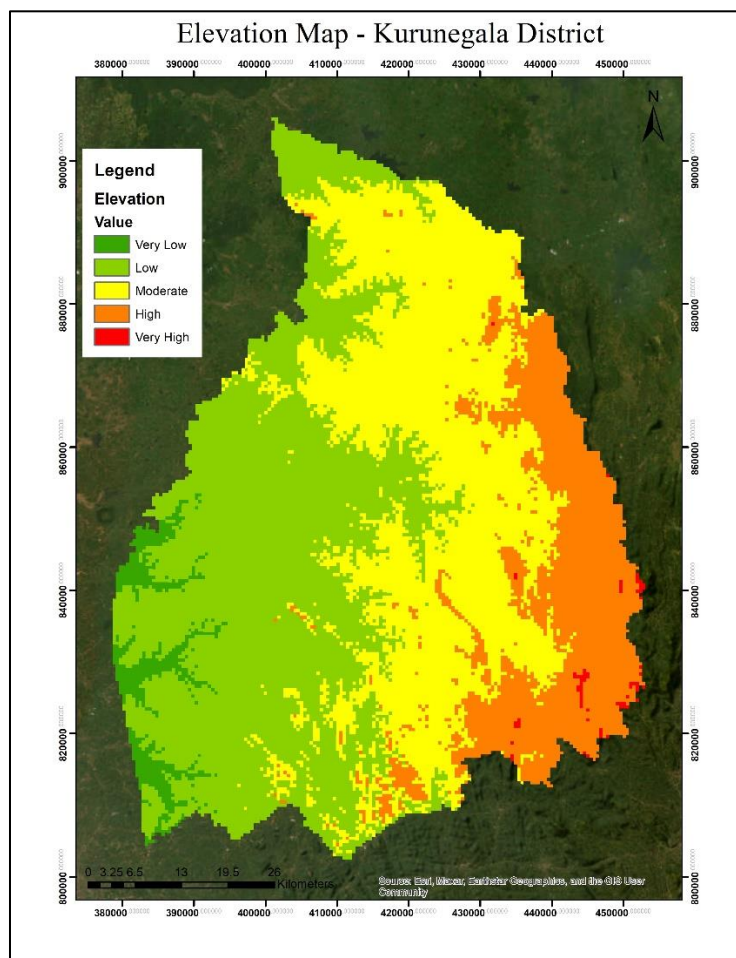


Figure 5 - Elevation Map

The map shows the elevation above sea level of the Kurunegala District in Sri Lanka. Higher elevations are represented by darker colors, while lower elevations are lighter. The legend indicates a range of elevation values, from "Very Low" to "Very High."

High elevations have lower groundwater recharge potential. This is because the water table is likely to be deeper at higher elevations, making it more difficult for rainwater to infiltrate and reach the aquifer. Additionally, mountainous areas often have steeper slopes, which can lead to faster runoff and less infiltration. Low elevations have higher groundwater recharge potential. The water table is closer to the surface at lower elevations, making it easier for rainwater to infiltrate and recharge the aquifer. Additionally, flatter areas tend to have less runoff and more infiltration.

- Moderate to high recharge potential: The central and northwestern parts of the district, with lower to medium elevations (light green and yellow), are likely to have moderate to high groundwater recharge potential.
- Moderate recharge potential: The southern and eastern parts, with medium elevations (light orange), may have moderate groundwater recharge potential.
- Low recharge potential: The northeastern and southwestern corners, with higher elevations (red), are likely to have low groundwater recharge potential.

Soil Classification

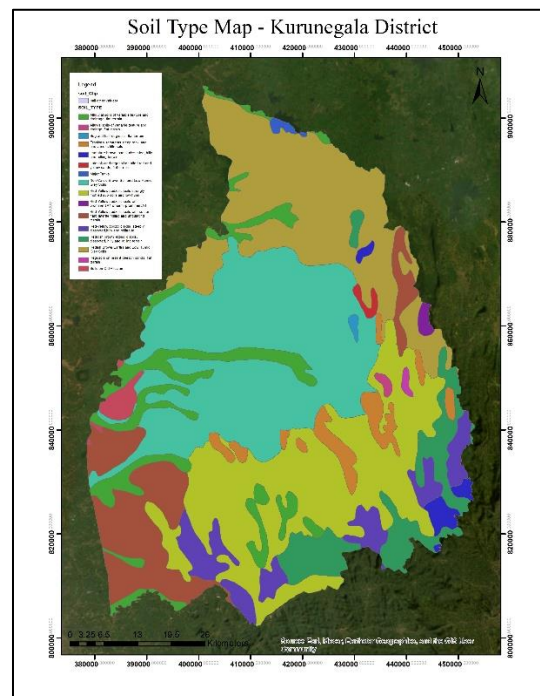


Figure 6 - Soil Map

Topography analysis

After doing the above steps I overlaid the above outputs to get the weighted topography for the further analysis using raster equation.

$$\text{Weighted_Topography} = (\text{Elevation_Raster} * \text{Weight_Elevation}) + (\text{Slope_Raster} * \text{Weight_Slope}) + (\text{Aspect_Raster} * \text{Weight_Aspect})$$

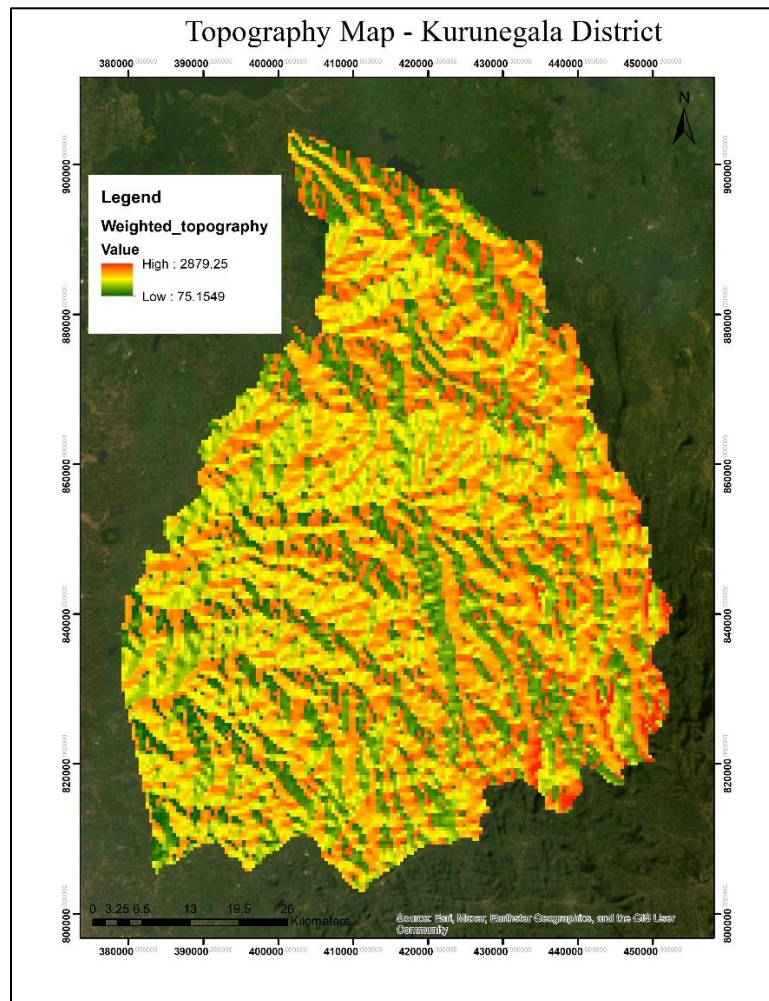


Figure 7 - Topography Map

- High recharge potential: The areas in light green and yellow, particularly in the central and northwestern parts of the district, have low to medium elevations and gentle slopes. These areas are likely to have good groundwater recharge potential. The valleys along the major rivers in the

central and southeastern parts of the district, indicated by the blue lines, may also have high recharge potential due to the presence of alluvial soils and flat topography.

- Moderate recharge potential: The areas in light orange, mainly in the southern and eastern parts of the district, have moderate elevations and slopes. These areas may have moderate groundwater recharge potential.
- Low recharge potential: The areas in red, concentrated in the northeastern and southwestern corners of the district, have higher elevations and steeper slopes. These areas are likely to have low groundwater recharge potential. The hilltops and mountainous areas in the northeastern and southeastern parts of the district are also likely to have low recharge potential due to their elevation and slope.

Deduru oya Basin

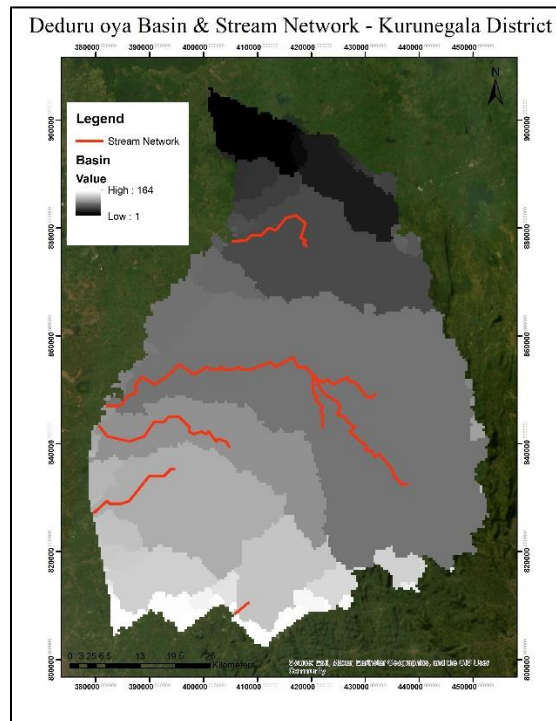


Figure 8 - Deduru oya Basin Distribution

- The central and northwestern corners of the district have high basin value because of that that area have high moisture.
- The eastern and southern parts of the district have moderate basin value.
- The northeastern and southwestern parts of the district have low basin value because of that that area have low moisture value.

3.5 Recharge estimation

Assessing groundwater recharge potential through spatial analysis involving factors like land cover, soil type, topography, and vegetation health is crucial for several reasons. Identifying areas with higher recharge potential aids in sustainable water resource management, especially in regions reliant on groundwater for domestic, agricultural, and industrial purposes. Understanding where recharge is more likely to occur allows for informed decision-making in land use planning, conservation efforts, and water resource allocation. By pinpointing these zones, authorities can implement targeted strategies such as infiltration pits, rainwater harvesting, or vegetation preservation to enhance groundwater replenishment, ensuring the long-term availability of this vital resource and mitigating the impacts of water scarcity and depletion in susceptible areas.

After doing all the above steps I overlayed the above results to get the Ground water recharge potential map. For that I used the below mentioned raster equation.

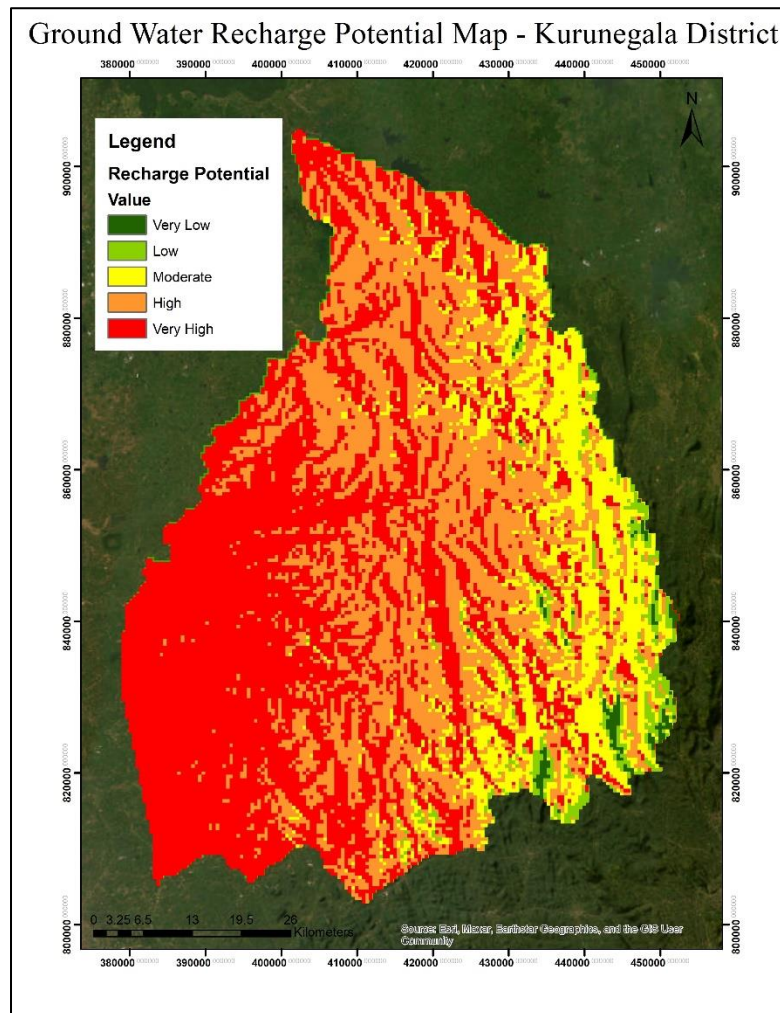


Figure 9 - Ground Water Recharge Potential Map

$$\text{Recharge_Potential} = (\text{LandCover_Raster} * \text{Weight_LandCover}) + (\text{SoilType_Raster} * \text{Weight_SoilType}) + (\text{Topography_Raster} * \text{Weight_Topography})$$

I. High Recharge Potential:

The central and northwestern corners of the district, particularly around the city of Kurunegala, display the highest recharge potential (dark orange and red). These areas likely have a combination of favorable factors like:

- Gentle slopes: Allowing for slower runoff and increased infiltration.
- Lower elevations: Placing the water table closer to the surface for easier recharge.
- Suitable soil types: Potentially sandy or well-drained soils facilitating infiltration.
- Vegetation cover: Possibly presence of forests or agricultural lands promoting infiltration.

II. Moderate Recharge Potential:

The eastern and southern parts of the district exhibit moderate recharge potential (light orange). These areas may have slightly steeper slopes, higher elevations, or less ideal soil types compared to the high potential zones.

III. Low Recharge Potential:

The northeastern and southwestern parts of the district show the lowest recharge potential (bright yellow and green). These areas likely have:

- Steep slopes: Leading to rapid runoff and minimal infiltration.
- Higher elevations: Placing the water table deeper and making recharge more challenging.
- Potentially rocky or less permeable soil types: Further hindering infiltration.

Notes.

Here I didn't consider the rainfall & precipitation data because there is ignorable difference in those values for Kurunegala district.

4. References

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For some analysis referred to ChatGPT and YouTube videos.